Nithik Vairamuthu

AI RESEARCHER

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https://nithik-vairamuthu-git-main-helios0007s-projects.vercel.app/

Architect and Computational designer working at the intersection of AI, generative systems and spatial intelligence. Currently engaged in developing AI driven design workflows at Institute for Advanced Architecture of Catalonia, focusing on programming, image segmentation pipelines and computer vision systems. Combines parametric tools with machine learning and hardware-based experimentation to build adaptive and data driven spatial systems. Interests lie in intelligent environments, computational design, and the future of habitat systems in extreme and extraterrestrial contexts.

Projects/ Research

LUNA: The Moon Base Dec 2023 -

Undergraduate thesis on exploring settlements in the lunar surface.

Chronoscape Oct 2025 -

Image generation workflow for assisting urban regeneration from memory as an input.

Retrograde Jan 2026 -

Façade feasibility analysis for green building conversion using image segmentation and regression model.

RE Pair Jan 2026 -

Image generation workflow for assisting urban regeneration from memory as an input.

Project Theia Mar 2026 -

Lunar environment analysis platform

Experience

Architectural Internship Dec 2022 ï May 2026

Iinfinity Architects, Chennai

Worked in Commercial, Residential and Project Documentation.

VAP Parametric workshop - VIT University, Vellore

Parametric modelling, fabrication and furniture design

Parametric workshop ï Tutor ï Mysore School of Architecture

Assisted in organizing and conducting a 3-day workshop.

Core Skills

AI and computer vision, including Python, PyTorch, OpenCV, Segment Anything Model (SAM), image segmentation, regression models, Stable Diffusion, and image generation workflows; computational design using Grasshopper and Python in Rhino; hardware prototyping with Arduino, ESP32, and sensor integration; Architecture and spatial design focused on architectural design, adaptive environments, and extreme or extraterrestrial habitats; and Strategic thinking with a design and systems-oriented approach.

Education

Institute for Advanced Architecture of Catalonia Oct 2025 ï

Masters in AI for Architecture and Built Environment

EADA Business School Oct 2025 ï

Business Strategist, Business Administration and Management, General

Vellore Institute of Technology, Vellore

Bachelors in Architecture (B.Arch)

Language

Tamil (native), English (Fluent)

Publication

Computational study of origami inspired geometries in lunar habitations Nov 2025

ZMECH Conference – VIT Vellore

Interests

AI assisted workflows, extreme habitats and extraterrestrial systems design, inter disciplinary collaboration.